

R&D Tax Credit Study

Project: Hudeman Slough Boat Ramp — Ecological Restoration (Job 338)

Project ID: P001

Prepared for: Gordon N. Ball, Inc.

Tax Year(s): 2025

1. Executive Summary

This report documents the Qualified Research Expenditures (QREs) and qualified research activities for project P001: Hudeman Slough Boat Ramp — Ecological Restoration (Job 338), conducted by Gordon N. Ball, Inc..

Business Component: New marine infrastructure system — boat ramp, dock structure, and turbidity/acoustic management system at Hudeman Slough tidal wetland

Problem Statement: The project required developing construction methods for installing precast concrete support piles and a dock/ramp structure in a tidal aquatic environment subject to strict California Department of Fish and Wildlife (CDFW) biological thresholds. Key technical challenges included: (1) determining the optimal pile driving sequence and hammer energy parameters to remain within acoustic impact thresholds for fish (dB SPL and dB SEL limits); (2) designing turbidity curtain configurations that would contain excavation-generated turbidity within regulatory limits under variable tidal hydrodynamic conditions; (3) engineering an effective sediment removal and stockpile management system for unconsolidated aquatic sediments; and (4) confirming pile bearing capacity in variable, saturated alluvial soils where standard penetration test data was insufficient.

Key Technical Uncertainty: At project initiation, the following technical uncertainties existed and could not be resolved from existing knowledge or standard construction practice: (1) Whether the proposed pile driving equipment selection and sequence would achieve the required pile bearing capacity while simultaneously remaining within the biological acoustic thresholds — the interaction between optimal driving energy for bearing capacity and maximum permissible acoustic levels was not determinable in advance; (2) Whether turbidity curtains could be configured to contain excavation turbidity under the site's specific tidal flow regime, given that turbidity plume behavior in a dynamic tidal wetland is highly site-specific and non-deterministic; (3) Whether the proposed BMP configurations for the Water Pollution Control Program (WPCP) would achieve the required effluent standards during sediment removal and stockpile operations; and (4) What the actual bearing capacity of the subsurface soils would be, and whether the pile foundation system design would achieve structural integrity with the in-situ soil conditions encountered during construction.

Outcome: All hypotheses were validated through iterative field experimentation. Pile installation was completed successfully within acoustic thresholds through adaptive hammer settings developed during construction. Turbidity curtain configurations were refined to achieve compliance throughout excavation phases. The completed boat ramp structure was accepted by the project owner and relevant regulatory agencies.

Metric	Value
Project	P001: Hudeman Slough Boat Ramp — Ecological Restoration (Job 338)
Company	Gordon N. Ball, Inc.
Active Tax Year(s)	2025
Total Project QREs (all active years)	\$168,513.65
Proportional Credit Attribution (all years)	\$10,110.82
Credit Method	ASC (Alternative Simplified Credit) — company-level

2. Company Background

2.1 Company Identity

Field	Value
Legal Name	Gordon N. Ball, Inc.
EIN	68-0350143
Entity Type	S-Corp
State of Incorporation	CA
States of Operation	CA
Industry	Heavy Civil and Environmental Construction
Tax Year Under Study	2025
Funded Research Exists	No
Is Startup (IRC §41(h))	No
Prior R&D Credit Claimed	No
Prior Form 6765 Years	—

2.2 Business Overview

Gordon N. Ball, Inc. is a California-based heavy civil and environmental construction contractor with decades of experience executing technically complex infrastructure projects. In tax year 2025, the company undertook four R&D-intensive projects: (1) the Hudeman Slough Boat Ramp ecological restoration (Job 338), involving novel pile installation techniques in a tidal aquatic environment; (2) the Stewart's Point Skaggs Springs Road slope stabilization and reconstruction (Job 350), involving development of an innovative soldier pile retaining wall system for an active landslide on a California state highway; (3) the East South Bay Salt Pond dual bridge construction (Job 351), involving novel pile foundation and CIP bridge deck construction methods in active tidal salt pond conditions with sensitive ecological constraints; and (4) the West South Bay Salt Pond rock slope protection and island construction (Job 356), involving development of RSP placement configurations, habitat island construction in tidal soils, and HTZ grading methods for ecological restoration. All four projects required iterative engineering problem-solving under conditions where the optimal construction methods could not be determined in advance from established practice.

2.3 Products & Services

- Heavy civil construction
- Marine and waterfront construction
- Ecological restoration and environmental remediation
- Highway and road construction
- Slope stabilization and geotechnical construction
- Water resource and drainage infrastructure

2.4 R&D Departments

- Field Engineering
- Project Management
- Environmental Compliance
- Geotechnical Engineering

3. Qualified Research Activities

The following table identifies the qualified research activities (QRAs) performed under IRC §41(d) for each tax year covered by this report. Only activities that meet all four parts of the IRS qualification test are included.

Tax Year 2025

Field	Detail
Tax Year	2025
Project ID	P001
Project Name	Hudeman Slough Boat Ramp — Ecological Restoration (Job 338)
Business Component	New marine infrastructure system — boat ramp, dock structure, and turbidity/acoustic management system at Hudeman Slough tidal wetland
Technical Objective	Design and construct a new boat ramp and dock structure at Hudeman Slough while developing novel pile installation techniques for soft aquatic soils, acoustic impact management methods for pile driving near sensitive fish habitat, and adaptive turbidity control systems for tidal wetland conditions.
Scientific Discipline	Acoustic Impact Hypothesis: Specific pile driving hammer settings and driving sequences can be selected to achieve required pile penetration while keeping acoustic levels below CDFW-prescribed dB SPL
Project Period	2025-01-01 – 2025-10-31
Status	Completed

3. Project Research Activities

Tax Year 2025 — P001: Hudeman Slough Boat Ramp — Ecological Restoration (Job 338)

IRC References: IRC §41(d)(1) — qualified research four-part test, IRC §41(d)(2)(B) — new business component, IRC §41(d)(4)(A) — funded research exclusion not applicable (public works contract; Gordon N. Ball bears all financial risk; design is owned by public agency, construction methodology IP retained by contractor), Treas. Reg. §1.41-4 — qualified research definition, Treas. Reg. §1.41-4(b)(2) — shrinkback: qualifying R&D phases identified at the individual work phase level

ii) Project Description

The "Hudeman Slough Boat Ramp — Ecological Restoration (Job 338)" project for Tax Year 2025 focused on designing and constructing a new boat ramp and dock structure at Hudeman Slough. This initiative aimed to develop innovative techniques for installing piles in soft aquatic soils, managing acoustic impacts during pile driving near sensitive fish habitats, and implementing adaptive turbidity control systems suitable for tidal wetland conditions. The project addressed several technical challenges, including determining the optimal pile driving sequence and hammer energy parameters to comply with acoustic impact thresholds for fish, designing turbidity curtain configurations to contain excavation-generated turbidity within regulatory limits under variable tidal conditions, engineering a sediment removal and stockpile management system for unconsolidated aquatic sediments, and confirming pile bearing capacity in variable, saturated alluvial soils where standard penetration test data was insufficient.

The project involved iterative field experimentation to validate all hypotheses. Successful pile installation was achieved within acoustic thresholds by developing adaptive hammer settings during construction. Turbidity curtain configurations were refined to maintain compliance throughout the excavation phases. Ultimately, the completed boat ramp structure met the stringent biological thresholds set by the California Department of Fish and Wildlife (CDFW) and was accepted by the project owner and relevant regulatory agencies.

iii) New or Improved Business Component

In the Tax Year 2025, the R&D project led to the creation of a new marine infrastructure system at the Hudeman Slough tidal wetland, comprising a boat ramp, dock structure, and a turbidity/acoustic management system. This innovative system was developed through rigorous iterative field experimentation, which validated all hypotheses. The successful completion of pile installation within acoustic thresholds was achieved by developing adaptive hammer settings during construction. The turbidity curtain configurations were refined to maintain compliance throughout the excavation phases, based on real-time turbidity monitoring and adjustments to BMP configurations. The project involved multiple pile driving attempts to determine the optimal combination of hammer energy and driving sequence, ensuring both acoustic compliance and design bearing capacity. Initial turbidity curtain configurations were modified in response to tidal flow patterns that differed from pre-construction models. The completed boat ramp structure was accepted by the project owner and relevant regulatory agencies, marking a significant advancement in marine infrastructure development.

iv) Elimination of Uncertainty

At the initiation of the project in Tax Year 2025, several technical uncertainties were identified that could not be resolved through existing knowledge or standard construction practices. The primary uncertainties included: determining whether the proposed pile driving equipment selection and sequence would achieve the necessary pile bearing capacity while adhering to biological acoustic thresholds; configuring turbidity curtains to effectively contain excavation turbidity within the site's specific tidal flow regime; ensuring that the proposed Best Management Practice (BMP) configurations for the Water Pollution Control Program (WPCP) would meet effluent standards during sediment removal and stockpile operations; and confirming the actual bearing capacity of the subsurface soils to ensure the pile foundation system's structural integrity with the in-situ soil conditions encountered during construction.

The research team systematically addressed these uncertainties through field testing and iterative experimentation. To resolve the uncertainty regarding pile driving parameters, the team conducted field tests to identify specific hammer settings and driving sequences that achieved the required pile penetration while maintaining acoustic levels below the California Department of Fish and Wildlife (CDFW) thresholds for Pacific Salmon and Steelhead. The effectiveness of turbidity curtains was evaluated under the dynamic tidal conditions of Hudeman Slough, leading to the identification of optimal deployment patterns that contained excavation-generated turbidity within the project boundary. Iterative testing of BMP configurations ensured compliance with California Construction General Permit turbidity effluent standards during sediment-disturbing activities. Finally, the actual in-situ bearing capacity of the foundation soils was confirmed through data collected during pile driving, validating that precast concrete piles installed with the selected hammer at specified blow counts developed the design bearing capacity in the site's saturated alluvial soils. Through these efforts, the research eliminated the technical uncertainties that existed at the project's inception.

v) Process of Experimentation

During the tax year 2025, Gordon N. Ball, Inc. engaged in a systematic process of experimentation to enhance construction methodologies and environmental management practices. This process involved developing and testing acoustic monitoring protocols with real-time feedback loops during pile driving, which were crucial for minimizing disturbances in fish habitats. The team iteratively deployed and adjusted turbidity curtain configurations based on measured turbidity readings to optimize sediment containment. The Water Pollution Control Plan (WPCP) was developed and modified in response to observed stormwater quality, ensuring compliance with environmental standards.

The experimentation extended to testing alternative excavation approaches and equipment configurations for effective sediment management. Engineering reviews were conducted at each phase to adapt construction methods based on observed site conditions. Specific phases included Crew Biological Training, where site-specific biological monitoring protocols were developed, and Contractor Survey, which involved iterative survey and field layout testing to optimize pile locations. Site Review and Vegetation Clearing phases focused on assessing site conditions and testing clearing methods to minimize soil disturbance. K-Rail Relocation and Demolition phases evaluated structural removal sequences to reduce water quality impacts. Sediment Excavation and Stockpile phases tested equipment types and dewatering methods to maintain turbidity control and stockpile stability. Support Pile Installation involved real-time adaptive pile driving, adjusting hammer energy and refusal criteria based on acoustic readings. BMP Installation and Removal tested silt curtain configurations for optimal turbidity containment, while WPCP Maintenance and Turbidity Curtain Operations involved iterative adjustments based on turbidity and stormwater quality measurements. This comprehensive approach ensured that construction activities were environmentally responsible and technically efficient.

vi) Technological in Nature

The research project at Hudeman Slough was deeply rooted in the principles of various engineering and scientific disciplines, demonstrating a robust application of technological knowledge. Geotechnical engineering principles were employed to design the pile foundation and assess bearing capacity, ensuring the stability and durability of the marine infrastructure system. Marine, coastal, and hydrological engineering principles were crucial in understanding and managing turbidity and tidal flow dynamics, which are essential for the effective integration of the boat ramp and dock within the aquatic environment. The project also incorporated environmental biology to evaluate acoustic impacts on fish and establish water quality thresholds, thereby aligning the development with ecological protection goals. Structural engineering expertise was applied in the design of precast concrete piles and ramp systems, contributing to the innovative construction of the new marine infrastructure. Collectively, these efforts were directed towards developing a new business component under IRC §41(d)(2)(B), resulting in the creation of a previously non-existent property that enhances both functionality and environmental stewardship.

vii) Resolution

The technical uncertainties in the Hudeman Slough Boat Ramp — Ecological Restoration project (Job 338) for Tax Year 2025 were resolved through a series of iterative field experiments and adaptive strategies. The team successfully validated all hypotheses by completing pile installation within acoustic thresholds using adaptive hammer settings developed during construction. This process required multiple attempts to determine the optimal combination of hammer energy and driving sequence that met both acoustic and structural requirements. Additionally, the effectiveness of turbidity curtains in this tidal environment was unknown prior to the project, necessitating modifications to initial configurations based on real-time tidal flow observations and turbidity monitoring data. Through these adjustments, compliance was achieved throughout the excavation phases. The research also addressed uncertainties regarding the BMP configurations, which were refined iteratively to meet regulatory standards. The successful completion and acceptance of the boat ramp structure by the project owner and relevant regulatory agencies marked the resolution of these uncertainties. The project ultimately demonstrated the importance of adaptive field experimentation in resolving technical challenges, although questions about the variability of tidal flow patterns and their impact on future projects remain open for further investigation.

viii) Prior Art & Literature Review

Standard marine construction practice and published literature on pile driving in open water environments did not provide adequate guidance for the combined acoustic compliance, turbidity management, and foundation design challenges present at this specific tidal wetland site. CDFW biological opinion thresholds for acoustic impacts and California Construction General Permit turbidity standards imposed constraints not addressed in standard marine construction references.

Project QRE & Credit Attribution

Category	Amount
Total Project QRE	\$168,513.65

4. Four-Part Test Analysis

Tax Year 2025 — P001: Hudeman Slough Boat Ramp — Ecological Restoration (Job 338)

Criterion	Justification	Result
Test 1 — Permitted Purpose	The research activities were performed for the purpose of developing a new business component under IRC §41(d)(2)(B): a new marine infrastructure system (boat ramp, dock, and ecological protection systems) at Hudeman Slough. The constructed improvements constitute new property that did not previously exist.	PASS
Test 2 — Technological in Nature	The research relied on principles of geotechnical engineering (pile foundation design and bearing capacity), marine/coastal/hydrological engineering (turbidity and tidal flow dynamics), environmental biology (acoustic impacts on fish, water quality thresholds), and structural engineering (precast concrete pile and ramp system design). The process of experimentation was fundamentally grounded in physical and engineering sciences.	PASS
Test 3 — Elimination of Uncertainty	The research eliminated multiple technical uncertainties: (1) optimal pile driving parameters for simultaneous structural and acoustic compliance were unknown before field testing; (2) turbidity curtain effectiveness in this specific tidal environment was unknown; (3) BMP configurations required iterative testing before achieving regulatory compliance; and (4) actual in-situ bearing capacity of the foundation soils was uncertain until pile driving data was collected.	PASS
Test 4 — Process of Experimentation	Gordon N. Ball, Inc. conducted a systematic process of experimentation through: (1) developing and testing acoustic monitoring protocols with real-time feedback loops during pile driving; (2) iteratively deploying and adjusting turbidity curtain configurations based on measured turbidity readings; (3) developing and modifying the WPCP based on observed stormwater quality; (4) testing alternative excavation approaches and equipment configurations for sediment management; and (5) conducting engineering reviews at each phase to adapt construction methods based on observed site conditions. The process was documented through contemporaneous job cost records with phase-level labor and equipment tracking.	PASS

5. Cost Identification Methodology

A. Wage Selection Method

Employee qualification percentages were determined via: Time Tracking.

B. Contractor Qualification

Contractor expenses were included. The 65% limitation rule was applied appropriately.

6A. Employee Wage Schedule

Only employees with time allocated to this project are shown.

Tax Year 2025

Employee	W-2 Wages	Qual %	Proj %	Qualified Wages	Activity
Victor Antonio Gomez	\$15,500.00	100.0%	100.0%	\$15,500.00	Direct Research
Jose Jesus Gil	\$14,800.00	100.0%	100.0%	\$14,800.00	Direct Research
Alejandro Aguilar	\$14,000.00	100.0%	100.0%	\$14,000.00	Direct Research
Oscar Diaz Infante	\$13,000.00	100.0%	100.0%	\$13,000.00	Direct Research
Flegido Eric Gil	\$10,000.00	100.0%	100.0%	\$10,000.00	Direct Research
Carlos Martinez-Zepeda	\$8,000.00	100.0%	100.0%	\$8,000.00	Direct Research
Adrian Guzman	\$7,000.00	100.0%	100.0%	\$7,000.00	Direct Research
Daniel Torres Franyutti	\$6,000.00	100.0%	100.0%	\$6,000.00	Direct Research
Joseph F. Lombardo	\$5,000.00	100.0%	100.0%	\$5,000.00	Direct Research
Tony Arthur Martindale	\$4,200.00	100.0%	100.0%	\$4,200.00	Direct Research
Cody W. Morrill	\$4,000.00	100.0%	100.0%	\$4,000.00	Direct Research
Mason M. Storm	\$13,000.00	100.0%	100.0%	\$13,000.00	Supervision
Field Crew — Job 338 Collective	\$27,050.43	100.0%	100.0%	\$27,050.43	Direct Research
TOTAL				\$141,550.43	

6B. Contractor Schedule

Tax Year 2025

Contractor	Description of Work	Amt Paid	Qual %	65%	QRE
BKF Engineers	Land survey, engineering design review, and construction support engineering services for Hudeman Slough Boat Ramp ecological restoration project. Work included contractor survey, alignment verification, and engineering consultation on pile installation approach.	\$11,925.00	100.0%	65%	\$7,751.25
JS Digital Design	Digital design services for Hudeman Slough Boat Ramp ecological restoration project (Job 338), Phase 001050 (Contractor Survey). Services sourced from Job 338 Job Cost All Detail Report — AP entry, Invoice 18161 dated 8/26/25.	\$1,534.00	100.0%	65%	\$997.10
TOTAL					\$8,748.35

6C. Supplies Schedule

Tax Year 2025

Description	Vendor	Amount	Unit Cost	Useful Life	QRE
Acoustic monitoring instrumentation and associated materials for pile driving acoustic impact assessment — hydrophones, data loggers, and calibration equipment deployed to measure and validate compliance with CDFW-prescribed dB SPL and dB SEL thresholds for Pacific Salmon and Steelhead during pile installation phase	Acoustic Equipment Vendors (multiple)	\$12,672.09	—	—	\$12,672.09
Best Management Practice (BMP) installation materials — turbidity curtain panels, floats, anchors, and hardware used for iterative testing of turbidity containment configurations in tidal zone; silt barrier panels and installation hardware for BMP experimentation under variable tidal flow conditions	Various Construction Materials Vendors	\$5,542.78	—	—	\$5,542.78
TOTAL					\$18,214.87

10. Credit Calculation & Attribution

The Alternative Simplified Credit (ASC) under IRC §41 is computed at the company level on Form 6765. Section 10A shows the full company-level ASC worksheet for each tax year this project was active. Section 10B shows this project's proportional attribution based on its share of total company QREs — for internal allocation purposes only.

10A. ASC Computation Worksheet — Tax Year 2025

Line Item	Calculation	Amount
Current Year QRE	Sum of all qualified expenditures	\$2,496,432.76
Prior Year 1 QRE	Year – 1 QRE	\$3,548,975.17
Prior Year 2 QRE	Year – 2 QRE	\$0.00
Prior Year 3 QRE	Year – 3 QRE	\$0.00
Average Prior 3 Years	(Year–1 + Year–2 + Year–3) ÷ 3	\$0.00
Base Amount	Average Prior 3 Years × 50%	\$0.00
Excess QRE	Current Year QRE – Base Amount	\$2,496,432.76
Credit Method Used	IRC §41 election	ASC
Credit Rate	IRC §41(c)(5) — Alternative Simplified Credit	6% (Startup ASC — IRC §41(c)(5)(B): only 1 of 3 prior years had QREs)
Federal R&D Credit	Current Year QRE × 6% (Startup ASC — IRC §41(c)(5)(B))	\$149,785.97

A-01 Startup ASC Rate Applied (6%): Fewer than 3 of the preceding 3 taxable years had qualified research expenditures. Per IRC §41(c)(5)(B), the credit equals 6% of current-year QREs (no base calculation). This is the correct treatment — the standard 14% excess-QRE formula requires QREs in all 3 prior years.

Qualified Small Business (IRC §41(h)): Does NOT qualify as QSB. QSB eligibility requires ≤\$5M gross receipts for the prior 5 years. Gordon N. Ball, Inc. is an established heavy civil contractor and is not expected to qualify as a QSB. Confirm gross receipts with tax counsel.

Carryforward Note: If the federal R&D credit exceeds the company's current-year regular tax liability (after applicable limitations), the unused credit may be carried back 1 year and carried forward up to 20 years under IRC §39. The computed credit above is the gross credit before the §280C election or carryforward analysis.

10B. This Project's Proportional Credit Attribution

The table below shows this project's proportional share of the company-level R&D credit for each year it was active. These figures are for internal cost allocation only — the IRS receives one Form 6765 per entity per year.

Tax Year	Project QRE	Company QRE	Project Share	Company Credit	Project Attribution
2025	\$168,513.65	\$2,496,432.76	6.8%	\$149,785.97	\$10,110.82
TOTAL / AVG	\$168,513.65	\$2,496,432.76	6.8%	\$149,785.97	\$10,110.82

Important: Proportional credit figures are for internal attribution only. IRC §41 credits are filed at the entity level on Form 6765 — you cannot file separate credits per project.

11. Supporting Documentation Index

The following documentation supports the qualified research expenditures claimed in this study and is retained on file per IRC §6001 record-keeping requirements.

Contractor Records

Document / Reference	Referenced In
BKF Engineers — 338 R&D; JOB COST ALL DETAIL 103125.pdf — AP entries, BKF Engineers invoices	§8 Contractor Schedule
JS Digital Design — 338 R&D; JOB COST ALL DETAIL 103125.pdf — AP entry, Invoice 18161 dated 8/26/25, JS Digital Design, Phase 001050 Materials	§8 Contractor Schedule

13. Research Methodology & Compliance Analysis

13.1 Funded Research Analysis (IRC §41(d)(4)(A))

All projects are public works construction contracts. While the projects are funded by public agencies (Caltrans / CA State Parks / similar), the contracts are construction performance contracts — not research contracts. Gordon N. Ball, Inc. bears all financial risk of construction failure (not reimbursed for cost overruns or failed approaches). The construction methodology, engineering solutions, and R&D activities are developed at Gordon N. Ball, Inc.'s expense and risk. This structure does not constitute 'funded research' under IRC §41(d)(4)(A). The analysis of funded research vs. at-risk construction is supported by Prop. Reg. §1.41-2(a)(1) and Treasury guidance on contract R&D.

13.2 Controlled Group Analysis (IRC §41(f)(1))

Based on available information, Gordon N. Ball, Inc. does not appear to be a member of a controlled group of corporations under IRC §41(f)(1) / §1563. This must be confirmed with company tax counsel and ownership records before filing.

13.6 Documentation Standards & Substantiation

Substantiation Method: Phase-level job cost accounting with contemporaneous payroll tracking via Spectrum (Dexter+Chaney) construction management system

Retention Policy: Project records including phase-level job cost reports, certified payrolls, subcontractor invoices, engineering submittals, and environmental monitoring data are retained per IRC §6001 requirements and California public works record retention standards (minimum 5 years).